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An assessment of municipal solid waste management practices and challenges in Mogadishu, Somalia

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The present study aimed to assess municipal solid waste management practices and challenges in Mogadishu, Somalia. The specific research objectives were to examine the state of municipal solid waste management practices, challenges, and strategies for improving practices and promoting best practices. Data were collected using a mixed-methods cross-sectional design from 273 respondents including waste management workers and residents. The findings reveal that current practices are largely ineffective, with 49.1% of respondents rating services as “poor”; 46.2% of households currently resort to roadside dumping due to a lack of formal infrastructure. Significant barriers include inadequate financial resources, insufficient institutional capacity, technological deficiencies, and a lack of skilled personnel (58.3%). Practical recommendations include the implementation of “pay-as-you-throw” policies to ensure financial sustainability, the establishment of small-scale community recycling hubs to reduce landfill pressure, and the formalization of private-sector partnerships through rigorous legislative oversight and standardized disposal sites.

KEYWORDS

attitudes, citizen perception, waste management, community practice, environment, municipal solid waste

1 Introduction

The issue of solid waste management has become a significant challenge in many African cities, including Mogadishu, the capital of Somalia (Godfrey et al., 2019). The growing population, rapid urbanization, and economic development in these regions have contributed to the increasing generation of municipal solid waste, leading to significant environmental and public health concerns (Assefa, 2017). According to estimates, in 2016 the global community generated approximately 2.01 billion metric tons of municipal solid waste annually, with an average of 0.74 kilograms produced per person each day (Ram et al., 2021). However, at least 33% of this waste was not managed in an environmentally responsible manner, contributing an estimated 1.6 billion tons of carbon dioxide emissions, which account for roughly 5% of global greenhouse gas outputs (Shahab et al., 2022). In response, the Paris Agreement of 2017 involved over 80 nations commit to reducing emissions and improving waste management practices (Wang and Stanisavljevic, 2019). Solid waste is broadly defined as unwanted or discarded materials and byproducts that are no longer needed by the original user (Omar et al., 2019). This includes a wide range of items, such as household garbage, institutional waste, street sweepings, construction

and demolition debris, and commercial waste, which are typically collected and managed by local authorities (Olukanni et al., 2020). Solid waste management (SWM) consists of various of strategies and practices for dealing with garbage at all stages of its life cycle, from generation throughout collection and disposal sites (Sinha et al., 2011). (Ajega and Genga, 2019). The primary objective of this research is to evaluate the existing framework of municipal solid waste management in Mogadishu to identify systemic gaps and propose sustainable interventions. Specifically, the study addresses current state, institutional, financial, and technical challenges of municipal solid waste management. To better understand the factors influencing citizen participation in municipal solid waste management practices, this study adopts the Theory of Planned Behavior (TPB) proposed by Ajzen (1991). The TPB is a widely recognized social psychological model that links beliefs to behavior, suggesting that an individual's intention to perform a given behavior is the most immediate determinant of that behavior (Hunter et al., 2019). Previous studies focused on general waste management; this research is the first to apply the Theory of Planned Behavior (TPB) to Mogadishu's specific socio-political context.

2 Research methodology

2.1 Study site

Mogadishu is the capital and largest city of Somalia. The city is situated in the Banadir region which is made up of seventeen districts and the estimated population is 2.5 million (AREA OF RETURN INFORMATION BROCHURE MOGADISHU, 2020). The city is the second-fastest growing globally, with an annual growth rate of 4% (Hassan et al., 2024). About 23% of this population are Internally Displaced People (IDP) who live in informal settlements, making access to basic sanitation and waste disposal facilities a problem in the city (Hassan et al., 2024). The study was carried out all 17 constituencies with (217,108) Households [Estimation, 2014 (UNICEF Somalia, 2014)]. The target population was the waste management workers; Environmental Cleaning private Companies, Mogadishu subdivision (county) sanitation directors, dumping sites operator, Mogadishu Municipality legal advisors, Directorate of Environment of the Office of the Prime minister, environmental experts, health workers, and waste generators.

2.2 Research design

This study utilized a descriptive, mixed-methods cross-sectional survey design (Trochim and Donnelly, 2001) to assess the role of municipal solid waste management practices and challenges in Mogadishu (Umer et al., 2019). This approach allowed for the integration of quantitative data from questionnaires and qualitative insights from interviews to provide a comprehensive assessment of the waste management landscape (Goering and Streiner, 1996).

2.3 Data types and sources

This study used primary data received from the targeted population through questionnaires, and interviews, while secondary data was obtained from earlier studies, publications, journals, and reports on solid waste management (Thomas, 2014).

2.4 Sampling methods and procedures

The purposive sampling technique was utilized for selecting specialized participants, leading to a distribution of 384 questionnaires. Of these, 273 were completed and returned, resulting in a robust 71.1% response rate. The target sample of 384 was not fully reached—a shortfall attributed to unavoidable logistical challenges like participant unavailability, non-response attrition, or outright refusal.

2.4.1 Pilot testing and qualitative detail

The questionnaire (Sections A: Demographics, B: Practices, C: Barriers) was pilot-tested with 30 residents from a non-sampled district to ensure reliability (Cronbach's alpha). Furthermore, 10 semi-structured interviews were conducted with key informants, including district sanitation officers and private managers. Qualitative data were analyzed using thematic analysis to identify recurring themes such as infrastructure deficits and financial constraints.

2.5 Methods of data analysis

Quantitative data were edited, coded, and analyzed using SPSS and Excel to generate descriptive statistics, including frequencies and percentages. Qualitative data were analyzed using thematic analysis, where interview transcripts were categorized into themes such as “infrastructure gaps,” “legislative oversight,” and “behavioral barriers” to complement and triangulate the quantitative findings.

2.6 Sample size

To determine the sample size, the Krejcie and Morgan formula was used (Krejcie and Morgan, 1970). For a target population of 217,108 households at a 95% confidence level, the target sample was 384. While the final achieved sample was 273 (response rate of 71.1%), this resulted in a margin of error of approximately 5.9%, which is slightly higher than the initial 5% target.

The mathematical application of the formula was as follows:

$$N = 217108$$

$$n = N / (1 + N(e^2)) = 217108 / (1 + 217108(0.1)^2)$$

$$n = 384$$

$$n = \text{Sample size}$$

$$N = \text{population size}$$

$$e = \text{level of significance}$$

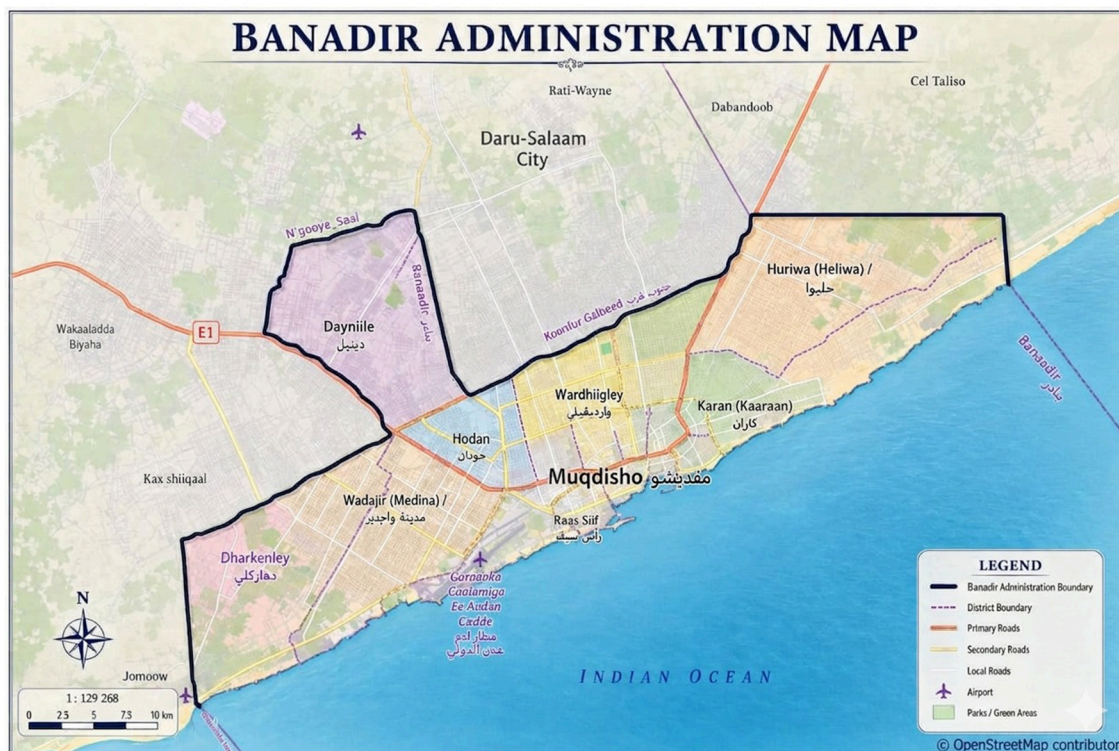


FIGURE 1
Map of Mogadishu City, Banadir region. Figure created by the authors using open-source map data from OpenStreetMap (ODbL).

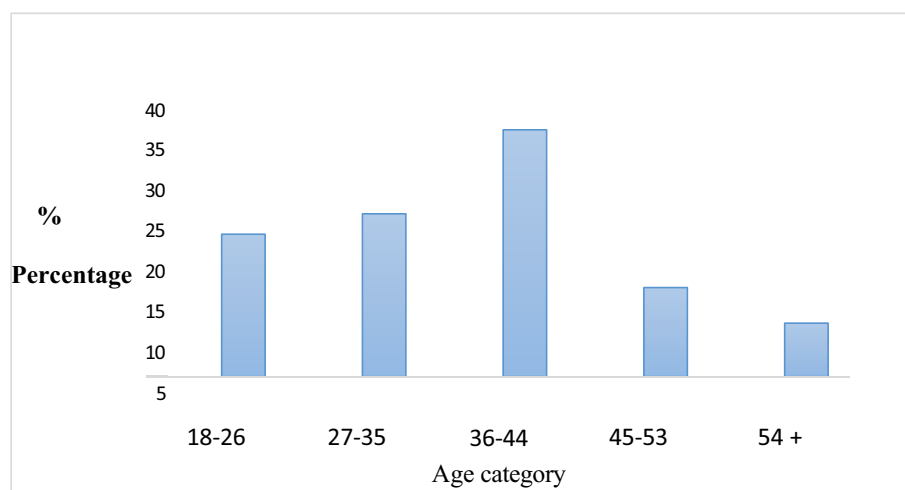


FIGURE 2
Age distribution of respondents.

3 Results and discussion

3.1 Response rate

Based on the primary data collection, the response rate was calculated. 273 of the 384 questionnaires were completed and returned, demonstrating a 71.1% response rate.

This response rate was satisfactory. According to Mugenda and Mugenda (2003), a response rate of 50% is

adequate for analysis and reporting; 60% is good, and 70% is very good (Mugo et al., 2018).

Figures 1, 2 indicates that the majority of respondents (35.5%) are in the age category 36–44 years, followed by 23.44% in the age category 27–35 years, and 20.5% in the 18–26 years category, with the fewest respondents in the 45 years and above category (Figure 3). Of these, 158 (58%) were male while 115 (42%) were female. The highest number of respondents (91, 33.3%) held a diploma, followed by 26.7% with a certificate; and 24.5%

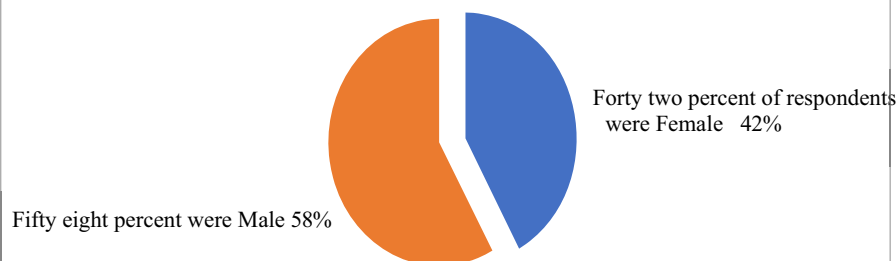


FIGURE 3
Gender distribution of respondents.

TABLE 1 Socio-demographic characteristics.

Variable	Category	Frequency	Percentage
Age	18–26	56	20.50
	27–35	64	23.44
	36–44	97	35.53
	45–53	35	12.82
	54 +	21	7.71
	Total	273	100.0
Gender	Female	115	42.0
	Male	158	58.0
	Total	273	100.0
Level of education	Master	18	6.6
	Bachelor degree	28	10.3
	Diploma	91	33.3
	Certificate	73	26.7
	Other	63	23.1
	Total	273	100.0
Work experience	0–2 years	69	25.3
	2–3 years	96	35.2
	4–5 years	57	20.8
	6–8 years	32	11.7
	9 + years	19	7.0
	Total	273	100.0

other qualifications. 16.9% had a bachelor's or master's degree combined as the highest academic achievement (Table 1). For work experience, results indicate that 35.2% of respondents had 2–3 years of work experience, followed by 25.3% with 0–2 years, 20.8% with 4–5 years, 11.7% with 6–8 years, and 7% with 9 + years of experience.

The metrics clarifying current operational constraints and public perception values are displayed systematically in Table 2,

with structural barriers further analyzed across variables in Table 3. A chi-square test was performed to examine the relationship between age and disposal behavior. Results showed that roadside dumping was significantly more common among residents aged 18–26 (67.9%) compared to those over 45 (38.1%), $\chi^2(4, N = 273) = 12.4, p = 0.015$. Additionally, an ANOVA test indicated significant differences in mean satisfaction ratings between districts ($F = 3.12, p < 0.05$).

3.2 The state of municipal solid waste management practices in mogadishu

This section presents findings from the study, which aimed to assess municipal solid waste management practices in Mogadishu.

Solid waste management in Mogadishu is undeniably in a critical state, marked by a pervasive lack of proper infrastructure and effective systems. The overwhelming majority of residents (49.1%) perceive solid waste management as “poor,” with an additional 20.9% rating it as fair. Qualitative insights reveal that this infrequency leads to “waste piling,” where residents feel forced to use roadside dumping (46.2%) as a survival mechanism. This mirrors findings in Garowe, where household waste often exceeds 1 kg per day, yet municipal services only reach a fraction of the population (Nour and Olusegun, 2024). This is further exacerbated by inefficient collection systems, with most respondents (46.5%) reporting weekly waste collection, and a significant portion (30–40%) of urban waste remaining uncollected, which directly impact less than half of the population. This aligns with broader trends cited by Wilson et al. (2013), who noted that 30–60% of urban solid waste in developing countries often goes uncollected, serving less than 50% of the population.

The challenges extend to the very foundations of waste disposal, as a staggering 86.1% of respondents highlight the absence of essential facilities at disposal sites, including compaction, proper enclosures, and drainage. This severe deficiency in formal disposal infrastructure directly contributes to the prevalent household waste

TABLE 2 State of municipal solid waste management practices in Mogadishu.

Variable	Category	Frequency	Percentage
Solid waste management rating	Excellent	14	5.2
	Very good	27	9.8
	Good	41	15.0
	Fair	57	20.9
	Poor	134	49.1
	Total	273	100.0
Solid waste collection	Daily	41	15.0
	Once every 3 days	48	17.6
	Once a week	127	46.5
	Twice a week	57	20.9
	Total	273	100.0
Effective mass media	Radio	131	48.0
	Television	39	14.3
	Newspaper	31	11.4
	Social media	72	26.3
	Total	273	100.0
Facilities at the disposal site	Yes	38	13.9
	No	235	86.1
	Total	273	100.0
Charge for waste collection	Yes	260	95.2
	No	13	4.8
	Total	273	100.0
Household waste disposal	Dustbin	25	9.2
	Roadside	126	46.2
	Empty space ear the house	93	34.0
	Inside the house	29	10.6
	Total	273	100.0
Current SWM practices	Sweeping	85	31.1
	Collection service	43	15.8
	Transporting	61	22.3
	Burying and burning	71	26.0
	Recycling and reuse	14	4.8
	Total	273	100.0
Effectiveness of private waste collectors	Yes	80	29.3
	No	193	70.7

disposal practices, where nearly half of residents (46.6%) resort to dumping waste at roadsides, and another 34.0% use empty spaces near their homes. This is despite the fact that a vast majority (95.2%) of Mogadishu residents are charged for waste collection. This aligns with Ritter (2019) who noted that Mogadishu, with its two million people, produces about 2,500 tons of waste daily. This high fee does not translate into improved services, underscoring a significant disconnect between payment and provision, and fostering negative attitudes toward the existing system, which undermines citizens' willingness to participate in proper waste management.

While the community expresses a clear preference for more organized collection methods, such as dustbins (35.9%) and

roadside containers (29.3%), these preferences, reflecting a desire for accessible infrastructure, are largely unmet. These observed behaviors, such as high rates of roadside dumping, align with the Theory of Planned Behavior's assertion that low perceived behavioral control, coupled with potentially negative attitudes and weak subjective norms, leads to reduced intention and undesirable behaviors. Qualitative interviews with district staff revealed that while 95.2% of respondents are charged for collection, the lack of standardized collection points forces residents to dump waste in "black spots" to avoid accumulating refuse at home. The World Bank website featured a statement, considered a historical baseline up to 2,000 that "it is common that 30–60% of all the urban solid

TABLE 3 Solid waste management practice challenges in Mogadishu (n=273).

STATEMENT	Responses (%)					Mean	Std. dev
	SDA	DA	NC	A	SA		
The collected domestic solid waste is dumped into the open dumping site	12.1	13.4	10.2	36.2	28.1	4.06	0.81
Mogadishu Municipality has allocated sufficient funds for the provision of effective SWM services in the city	28.6	38.3	8.7	12.5	11.9	2.98	0.74
The personnel involved Mogadishu's SWM services have sufficient qualification and skills in the field.	18.4	39.9	20.4	13.1	8.2	2.24	0.98
Waste operator companies have the necessary capacity, and equipment of SWM services in the city	13.3	35.1	24.2	17.5	9.9	2.02	0.78
Mogadishu Municipality does not provide clear sanitation rules and regulations	7.1	9.7	30.6	37.3	15.3	2.65	0.73
Mogadishu has sufficient landfills used for Solid waste management	25.8	42.3	15.3	10.0	6.6	2.26	1.35

waste in developing countries is uncollected and less than 50% of the population is served” (Global Waste Management Outlook, 2016).

Efforts to raise awareness through mass media, particularly radio (48.0%), offer a promising avenue for communicating community expectations and shaping positive subjective norms regarding waste disposal. However, effective communication must be coupled with tangible improvements in infrastructure and collection services to empower residents to adopt more responsible waste management practices and mitigate the severe health and environmental risks posed by the current situation (Olukanni et al., 2020). The fact that 86.1% of lack disposal facilities indicates extremely low Perceived behavioral control (PBC), as residents feel they lack the means to manage waste correctly even if they intend to (Guichard et al., 2023). UN-HABITAT (2012), which estimated that 1,500 tons per day of solid waste are generated in the urban areas of Somalia, with uncontrolled disposal posing serious health risks and of abuse of the right to live in a healthy environment.

Challenges identified include the widespread practice of dumping collected domestic solid waste in open yards (36.2% agreed, 28.1% strongly agreed). Respondents largely disagreed that Mogadishu Municipality allocates sufficient funds for effective SWM, including reduction, recycling, and recovery programs. A significant majority (58.3%) also indicated that SWM personnel lack necessary qualifications and skills.

Additionally, more than 50% of respondents disagreed with the claim that waste operator companies have the necessary capacity and equipment. There is also a shortage of landfills, with 42.3% disagreeing and 25.8% strongly disagreeing that Mogadishu has enough landfills. Experts providing qualitative insights highlighted that the “open dumping” practiced in Mogadishu (supported by 64.3% agreement in the survey) can lead to significant groundwater contamination risks (Kanmani and Gandhimathi, 2013). Supportive material and informational conditions are crucial for a strong sense of performance ability in waste management (Guichard et al., 2023). Feedback from municipal directors highlighted that the lack of “designated disposal sites” often results in the use of unauthorized dumpsites near residential areas. Recent studies in Mogadishu’s Wadajir and Dharkeynley districts confirm that these improper dumpsites are directly linked to health risks, including increased malaria transmission due to breeding sites for disease vectors, as a significant association was

TABLE 4 Sampling table.

Participant group	Sampling technique	Achieved sample (n)
Residents	Systematic (Every 5 th house)	263
Waste workers & experts	Purposive	10
Total		273

found between improper waste disposal and malaria transmission (Mohamed et al., 2023a). Furthermore, a lack of waste management infrastructure in Banadir contributes to marine pollution, with over 2,70,000 kg of plastic improperly managed daily in coastal regions (Mohamed et al., 2023b). Waste recycling is not a priority for most municipalities, often due to a lack of knowledge about recycling and associated opportunities (Soares and Fernandes, 2019).

Applying the Theory of Planned Behavior clarifies the gap between resident intentions and actual practices in Mogadishu. As shown in Table 4, the high rate of roadside dumping (46.2%) is not merely a lack of awareness but a reflection of critically low Perceived Behavioral Control. When 86.1% of the population lacks access to proper facilities, the ‘control beliefs’ are dominated by external constraints rather than personal choice (Guichard et al., 2023; Table 5).

Furthermore, the negative attitudes fostered by poor service delivery (49.1% ‘poor’ rating) undermine the perceived benefit of complying with municipal regulations. However, the strong reliance on Radio (48.0%) for information provides a strategic ‘subjective norm’ lever; by using radio to broadcast community-led successes and social expectations, the municipality can foster a sense of collective responsibility that may eventually improve behavioral intentions once the infrastructure gaps are addressed (Hunter et al., 2019; Olukanni et al., 2020).

3.3 Strategies for proper solid waste management in mogadishu

The respondents were further told to rate strategies for promoting public participation in solid waste management

TABLE 5 Mapping of Mogadishu SWM Findings to TPB Constructs.

TPB Construct	Key Research Finding	Behavioral Implications
Attitude	49.1% of respondents rate current services as “poor” [Table 2].	Negative evaluation of the service leads to low motivation to participate in formal systems or pay fees.
Subjective Norms	48.0% of residents identify Radio as the most effective mass media [Table 2].	Significant potential exists to shape social norms and community expectations through targeted radio outreach.
Perceived Behavioral Control	86.1% of respondents report a lack of disposal facilities; 46.2% resort to roadside dumping [Table 2].	Residents feel they lack the means to manage waste correctly, even if they intend to do so, due to structural barriers

TABLE 6 Response rates.

Categories of questionnaire	Frequency	Percentage
Number returned	273	71.1
Number not returned	111	28.9
Total supplied	384	100.0

practices in Mogadishu, feedbacks obtained were presented in Table 6 below.

Respondents identified several strategies to promote public participation, aligning with TPB constructs feedbacks obtained for enhancing municipal performance parameters are presented in Table 7 below. Weekly awareness-raising programs (36.2% agreed, 29.3% strongly agreed) are crucial for improving knowledge and attitudes, thereby fostering more favorable attitudes and strengthening positive subjective norms. Distributing necessary SWM facilities to households and markets (39.7% agreed, 8.6% strongly agreed) directly enhances perceived behavioral control (PBC) by providing the tools for proper waste collection, sorting, and disposal. Developing and properly implementing enhanced waste management regulations (30.4% agreed, 30.7% strongly agreed) also improves PBC by clarifying expectations and reinforcing subjective norms by establishing clear societal standards. Adapting ‘pay-as-you-throw’ policies can influence attitudes toward waste generation and disposal by creating financial incentives (Knickmeyer, 2019). Supporting small-scale recycling industries is also seen as a sustainable solution. These findings align with recent studies on “The Blue Economy” and urban resilience, suggesting that without legislative enforcement and “pay-as-you-throw” models (supported by 44.8% of respondents), the financial gap in Mogadishu’s waste sector will continue to widen. This mirrors challenges seen in other rapidly urbanizing East African hubs like Nairobi, where informal settlements often outpace formal infrastructure (Santos et al., 2017). Principle, the Environmental Management Act, provides citizens with the right to a clean environment, and enabling them to understand their role in maintaining it and to compel polluters to pay for environmental damage.

The findings align with broader trends in developing nations where rapid urbanization outpaces municipal capacity (Ahmad et al., 2025; Moni et al., 2024). The 58.3% skill gap in

Mogadishu is a critical hurdle that prevents 3transition from “open dumping” to “integrated resource recovery” (Nour and Olusegun, 2024). Like other African cities, Mogadishu struggles with a lack of community engagement and limited resources. To address this, the implementation of “4R” strategies (reduce, reuse, recycle, recover) supported by public-private partnerships is essential for long-term sustainability (Jacob and Luyiga, 2024).

4 Limitation of the study

- 1-This study has several limitations: the cross-sectional design prevents drawing causal inferences; the achieved sample ($n = 273$) was lower than the target ($n = 384$), increasing the margin of error to 5.9%; and the use of purposive sampling for experts may limit the generalizability of the qualitative findings to other regions in Mogadishu.
- 2-Respondents might provide socially desirable answers rather than reflecting their true waste disposal habits.
- 3-A lack of official municipal records made it difficult to compare primary findings with historical secondary data.

5 Conclusion

The study assessed the current challenges and strategies of solid waste management practices in Mogadishu municipality. The majority (49.1%) of respondents indicated poor solid waste management in Mogadishu. The study identified several key barriers to effective municipal solid waste management in Mogadishu. These included a lack of capacity, facilities, and equipment among private waste collection firms for properly managing waste collection, transfer, and disposal. Additionally, personnel involved in solid waste management lacked the necessary qualifications and skills to effectively carry out waste collection, transfer, and disposal operations. There was also a reported shortage of landfill infrastructure for solid waste disposal. Furthermore, respondents indicated that financial allocation was inadequate to support effective solid waste management services, including waste reduction, recycling, and recovery initiatives. The study concluded that the most effective strategies for improving municipal solid waste management in Mogadishu include charging waste management fees to all community members;

TABLE 7 Strategies for Effective Solid Waste Management Practices in Mogadishu.

STATEMENT	Responses (%)					Mean	Std. dev
	SDA	DA	NC	A	SA		
Raising awareness programs to improve knowledge and attitude toward public participation in SWM in Mogadishu	5.4	9.3	19.8	36.2	29.3	3.03	1.01
Distribution necessary facilities of SWM to the households and markets reduces waste dumped to the undesignated areas.	16.3	21.1	14.3	39.7	8.6	2.08	0.81
Building enough sanitary landfills enhances proper SWM and environmental protection in Mogadishu.	3.5	5.2	8.7	24.3	58.3	4.18	1.26
Adapting a pay as you throw policy and changing fees and tax on waste generators to create funds for SWM and reduce the amount of waste generated.	17.4	21.5	16.3	24.5	20.3	2.05	0.51
Developing waste management rules and regulations, and implementing them properly limit Solid waste in Mogadishu	8.2	19.2	11.5	30.4	30.7	3.52	0.65
Backing the small-scale recycling industries is a durable solution to waste management in Mogadishu.	4.7	11.5	15.3	48.2	20.3	2.47	0.91

conducting weekly public awareness campaigns by the Mogadishu municipality on public health, solid waste management, and environmental issues; developing and properly implementing enhanced waste management regulations; and supporting local firms to reduce waste while establishing standardized disposal sites.

6 Recommendations

Strengthen policy oversight to establish performance-based contracts for private cleaning companies to address the 49.1% “poor” service rating currently reported by residents.

Financial reforms to implement “Pay-As-You-Throw” models for the 95.2% of households already paying for services, ensuring fees are tied to service quality and waste reduction.

Infrastructure investment to develop decentralized community recycling hubs to eliminate the 46.2% roadside dumping rate and the 86.1% shortage of proper disposal facilities.

Targeted public awareness to launch radio-based education campaigns (preferred by 48.0% of participants).

Capacity building to establish technical training programs to bridge the 58.3% skill gap identified among current waste management personnel.

Environmental protection to transition from open dumping to sanitary landfills to mitigate the impact of the 2,500 tons of daily waste currently polluting the coastline.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by National Commission on higher education. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

AM: Data curation, Methodology, Conceptualization, Formal analysis, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. SO: Data curation, Methodology, Conceptualization, Validation, Visualization, Software, Writing – review & editing. AG: Data curation, Methodology, Conceptualization, Formal analysis, Validation, Visualization, Writing – review & editing.

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Conflict of interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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